

12. A Precise Quarterback The number of passes completed and the total number of passing yards recorded for the Los Angeles Chargers quarterback, Philip Rivers for each of the 15 regular season games that he played in the fall of 2017¹³ were used to calculate the average number of yards per pass for each game.

Week	Completions	Yards per Pass	Week	Completions	Yards per Pass
1	28	13.8	9	17	12.5
2	22	13.2	11	15	12.2
3	20	11.4	12	25	10.7
4	18	17.7	13	21	12.3
5	31	11.1	14	22	15.8
6	27	16.1	15	20	11.8
7	20	12.6	16	31	10.7
8	21	17.2	17	22	8.7

- Calculate the mean and standard deviation of the number of completions and the number of yards per pass.
- Find a 95% confidence interval estimate for the variance of the number of completions. Why would you prefer a small variability in the number of completed passes?
- Find a 95% confidence interval for σ^2 , the variance of the yards per pass. Use these results to find a 95% confidence interval for σ , the standard deviation of the yards per pass.
- Test whether the standard deviation of the yards per pass for this quarterback differs from $\sigma = 4$ with $\alpha = .05$.

$$a. \quad \bar{x} = 22.5 \quad s_x = 4.7046$$

$$\bar{y} = 12.9875 \quad s_y = 2.531$$

$$b. \quad CI_{s_x^2} = \left(\frac{(n-1)s_x^2}{\chi^2_{(n-1), \frac{\alpha}{2}}}, \frac{(n-1)s_x^2}{\chi^2_{(n-1), (1-\frac{\alpha}{2})}} \right)$$

$$= \left(\frac{(16-1) \cdot 4.7046^2}{\chi^2_{15, 0.025}}, \frac{(16-1) \cdot 4.7046^2}{\chi^2_{15, 0.975}} \right) \approx (12.078, 53.018)$$

$$c. \quad CI_{s_y^2} = \left(\frac{(16-1) \cdot 2.531^2}{\chi^2_{15, 0.025}}, \frac{(16-1) \cdot 2.531^2}{\chi^2_{15, 0.975}} \right) \approx (3.496, 15.345)$$

$$CI_{s_y} = (1.86976, 3.91727)$$

$$d. \quad H_0: \sigma^2 = \sigma_0^2 = 4^2, \quad H_a: \sigma^2 = \sigma_0^2 \neq 4^2$$

$$\alpha = 0.05$$

$$\chi^2_{STAT} = \frac{(n-1)s_y^2}{\sigma_0^2} \sim \chi^2_{15}$$

$$\chi^2* = \frac{(16-1) \cdot 2.531^2}{4} \approx 6.02559$$

$$\chi^2_{15, 0.975} = 6.262$$

$$\chi^2_{15, 0.025} = 27.4884$$

$$\text{rejection region: } \{ \chi^2 > 27.4884 \text{ or } \chi^2 < 6.262 \}$$

$$\because \chi^2* = 6.02559 < 6.262$$

$$\Rightarrow \text{reject } H_0.$$